

Quantum Channels: Key Gotchas and Equivalence Statements

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Purpose

This document collects concise, high-impact facts about **quantum channels** (CPTP maps). Each statement is phrased as an equivalence, implication, or common intuition trap, useful for quick conceptual understanding and exams.

Basic Properties

- A quantum channel is a **completely positive trace-preserving (CPTP)** map.
- Trace-preservation ensures probabilities are normalized:

$$\text{Tr}[\mathcal{E}(\rho)] = 1 \quad \forall \rho.$$

- Complete positivity ensures the map remains positive even when acting on a part of a larger entangled system.
- Every quantum channel has a **Kraus representation**:

$$\mathcal{E}(\rho) = \sum_i K_i \rho K_i^\dagger, \quad \sum_i K_i^\dagger K_i = I.$$

- The Kraus representation is **not unique**; there exist infinitely many sets of Kraus operators for the same channel.

Types of Channels

- **Unitary channels**: $\mathcal{E}(\rho) = U\rho U^\dagger$ Reversible; entropy-preserving.
- **Depolarizing channel**: $\rho \mapsto (1-p)\rho + p\frac{I}{d}$ Introduces white noise; completely destroys information at $p = 1$.
- **Amplitude damping**: models energy loss (e.g., spontaneous emission).
- **Phase damping / dephasing**: destroys off-diagonal coherences, preserves populations.

Gotchas and Intuition Traps

- A channel can increase entropy but **cannot decrease it** in a closed system unless acting on a subsystem.
- A unitary channel does **not create entanglement** if the input is separable.
- A channel acting locally can **never increase entanglement** between two subsystems (LOCC principle).
- The Choi matrix of a channel is positive semidefinite *iff* the channel is completely positive.
- The action on the maximally mixed state gives insight into “noisiness”:

$$\mathcal{E}(I/d) = I/d \quad \Rightarrow \text{unital channel.}$$

- Depolarizing channels are unital; amplitude damping channels are non-unital.
- Composition of two CPTP maps is CPTP, but the inverse of a CPTP map may not exist (non-unitary channels are irreversible).
- Quantum channels can be simulated by unitary evolution on a larger system (Stinespring dilation theorem).
- Classical channels are a special case of quantum channels (diagonal density matrices).
- Kraus operators can be chosen to be orthogonal in Hilbert-Schmidt sense, but this is not necessary.
- A channel may preserve populations but destroy coherences (pure dephasing).
- Entanglement-breaking channels map *any input state* to a separable output.
- Not all positive maps are completely positive (e.g., transposition).
- The set of all quantum channels is convex: convex combinations of CPTP maps are CPTP.
- The identity channel is the only channel that preserves all states exactly.

One-Line Memory Triggers

- “CPTP $\Rightarrow$ Kraus decomposition exists.”
- “Unitary channel $\Rightarrow$ reversible, no entropy change.”
- “Entanglement-breaking $\Rightarrow$ all outputs separable.”
- “Depolarizing channel $\Rightarrow$ white noise, unital.”
- “Amplitude damping $\Rightarrow$ non-unital, irreversible.”
- “Complete positivity is stronger than positivity.”
- “Kraus representation is not unique.”
- “Local channel cannot increase entanglement.”
- “Convex combination of CPTP maps is CPTP.”